

5. The table contains information about three stars in the Milky Way galaxy.

Star	Surface temperature (K)	Radius (m)	Mass (kg)
star 1	2290	8.14×10^7	1.77×10^{29}
Sun	5800	6.96×10^8	1.99×10^{30}
star 2	210 000	5.57×10^8	5.69×10^{31}

- (a) Compare the surface temperature of star 2 with the surface temperature of star 1 in terms of orders of magnitude.

2

Space for working and answer

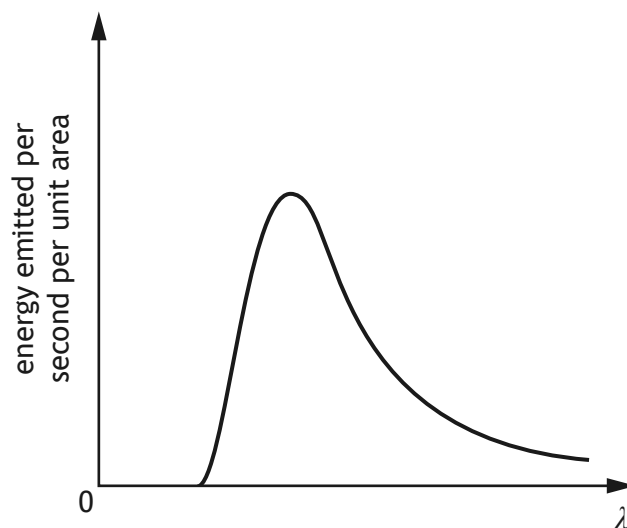


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5. (continued)

- (b) Stars emit radiation with a range of wavelengths.

The graph shows how the energy emitted per second per unit area varies with the wavelength λ of emitted radiation from the Sun.



- (i) On the graph above, add a line to show how the energy emitted per second per unit area varies with the wavelength of emitted radiation λ from star 2.

2

(An additional graph, if required, can be found on *page 51*.)

- (ii) Explain why star 1 appears more red than the Sun.

1

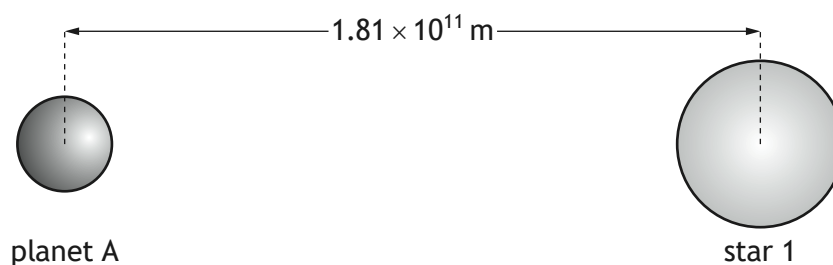
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5. (continued)

- (c) (i) Exoplanets are planets that are outside our Solar System. Star 1 is orbited by a number of exoplanets.

One exoplanet, planet A, has a mean orbital radius of 1.81×10^{11} m and a mass of 6.09×10^{24} kg.



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The mass of star 1 is 1.77×10^{29} kg.

Calculate the gravitational force between planet A and star 1.

Space for working and answer

3

5. (c) (continued)

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- (ii) A second exoplanet, planet B, has three times the mass of planet A.
The mean orbital radius of planet B is double the mean orbital radius of planet A.
State how the gravitational force of attraction between planet B and star 1 compares to the gravitational force of attraction between planet A and star 1.
Justify your answer.

3

- (iii) The exoplanets orbiting star 1 were discovered using the 'transit method'. The brightness of the light from the star is recorded. A dip in brightness is recorded during the time it takes the exoplanet to pass between the star and an observer.

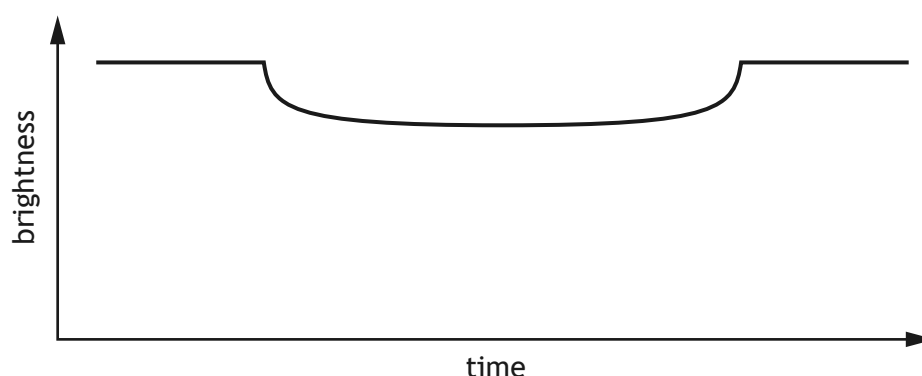
The table gives information about the exoplanets, planet A and planet B.

Exoplanet	Radius of exoplanet (m)	Mean orbital radius (m)	Orbital period (s)
planet A	7.1×10^6	1.81×10^{11}	1.4×10^8
planet B	5.0×10^6	3.62×10^{11}	4.0×10^8

The graph shows how the brightness varies with time as planet B passes between the star and the observer.

Add a line to the graph to show how the brightness varies with time as planet A passes between the star and the observer.

2



(An additional graph, if required, can be found on page 51.)

